Student Achievement & Equity Analytics Dashboard

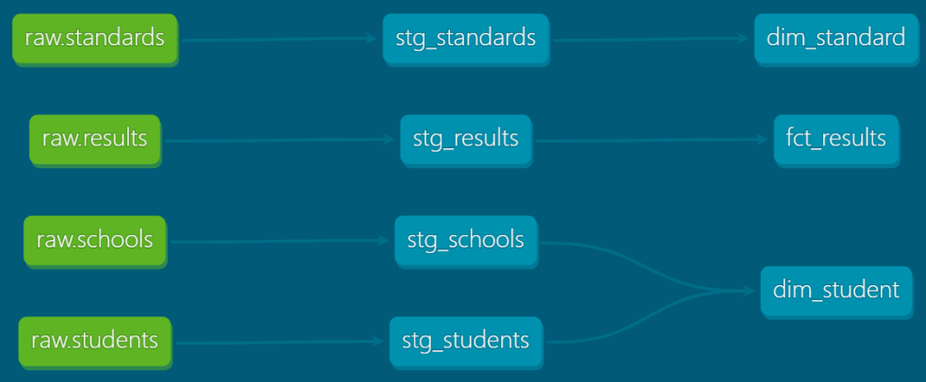
Goal:
Provide insights into student academic performance across subjects, schools, and socio-economic contexts, enabling analysis of achievement trends and equity gaps.

Data Pipeline:
Synthetic data is ingested into **Snowflake** from CSV sources, transformed using **dbt** into a structured star schema (staging and marts layers), and visualised in **Power BI** using an import model.

Description:
The dashboard is structured into three layers, enabling both high-level insights and deep exploration:

- Student Performance Overview,
- Performance by Socio-Economic Decile,
- and Subject-Level Analysis.

Data Lineage (DAG) from dbt:



Student Performance Overview

400	1.49	80.5%	20.8%
Total Results	Average Score	Pass Rate	Excellence Rate

Subject

All

School Region

All

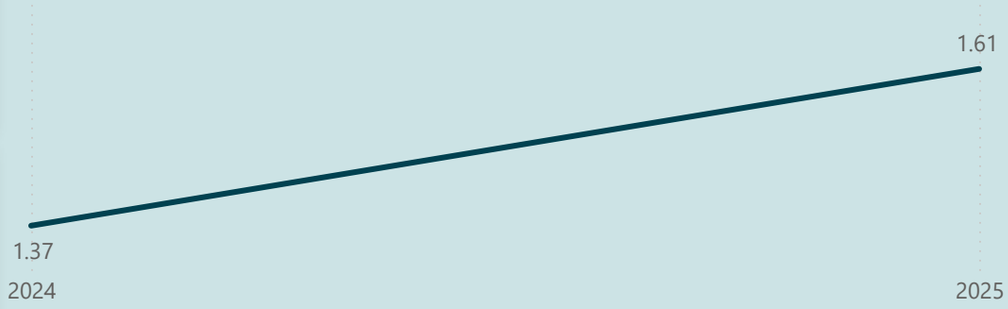
Decile

1 10

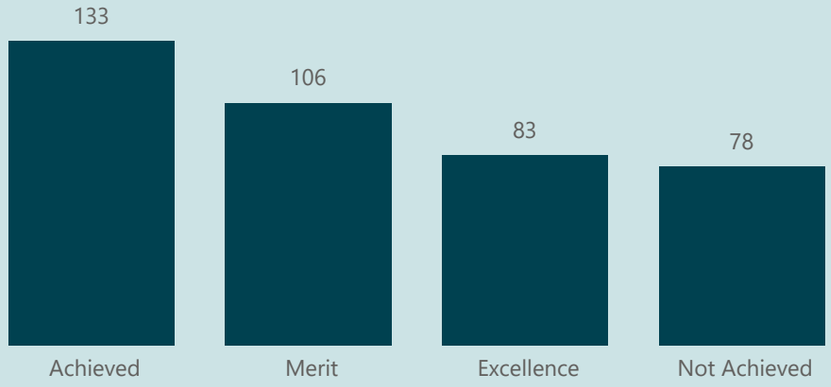
Year

2024 2025

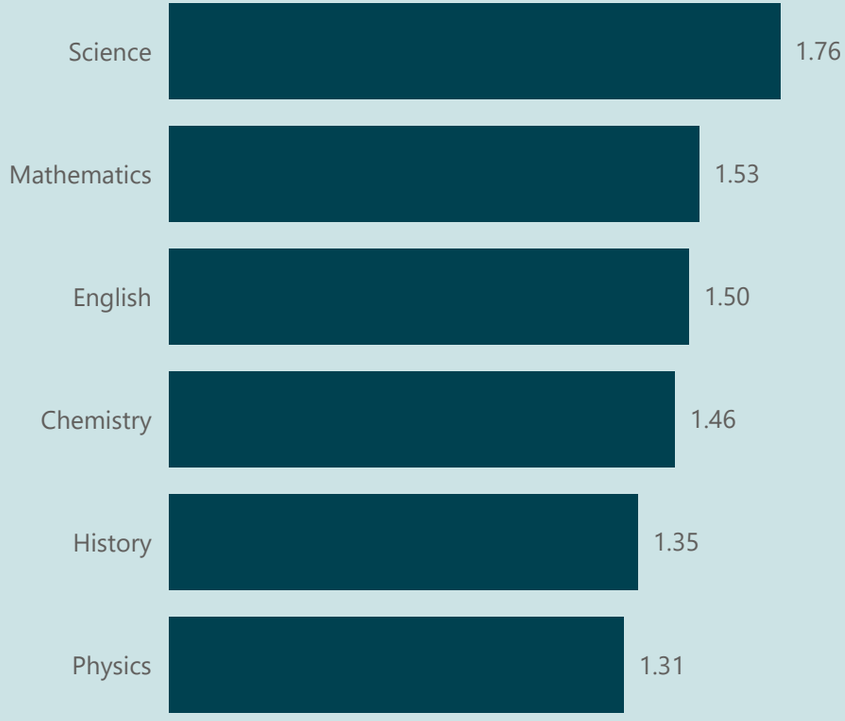
Results Trend (Average Score by Year)



Results Distribution



Subject Performance (Average Score by Subject)



Performance by Socio-Economic Decile

400	1.49	80.5%	20.8%
Total Results	Average Score	Pass Rate	Excellence Rate

Subject

All

School Region

All

Decile

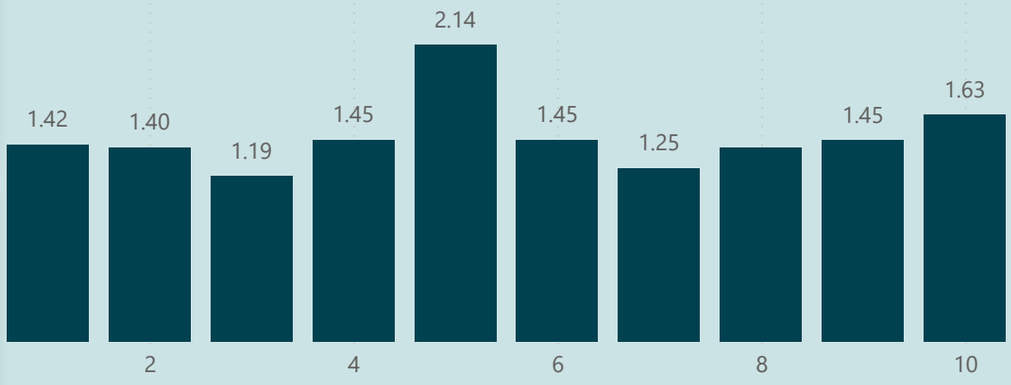
1 10

Year

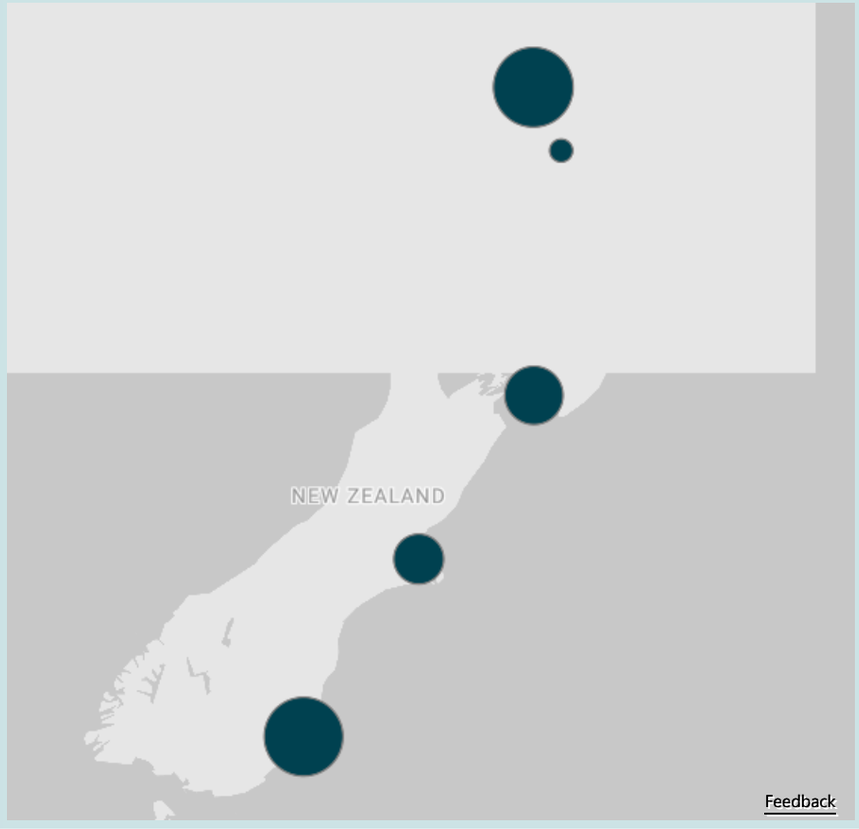
2024 2025

School Performance (Top 5 by Average Score)

Decile Analysis



Region Analysis



Feedback

Subject-Level Analysis

400	1.49	80.5%	20.8%
Total Results	Average Score	Pass Rate	Excellence Rate

Subject

All

School Region

All

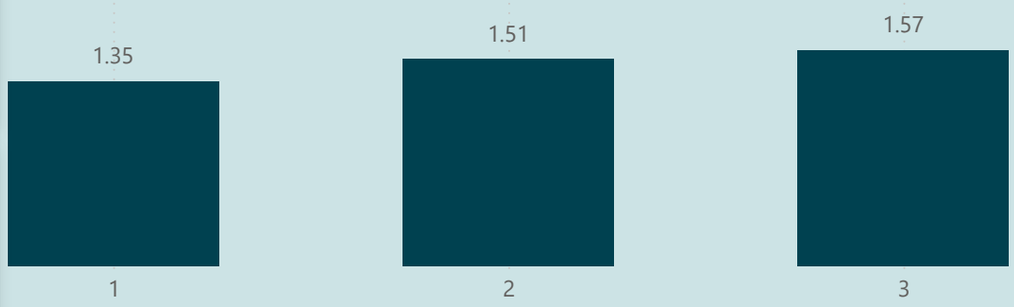
Decile

1 10

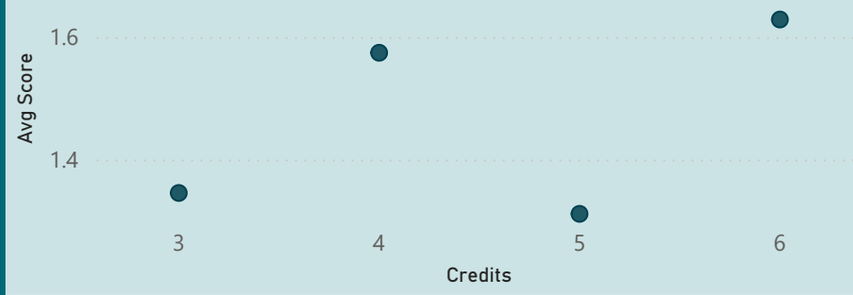
Year

2024 2025

Level Analysis



Credits vs Performance



Results Table

Subject	School Name	Total Results	Avg Score	Pass Rate	Excellence Rate
Chemistry	Central Tech	10	1.70	70.0%	40.0%
Chemistry	Eastview College	14	1.50	78.6%	28.6%
Chemistry	Greenfield School	6	1.00	66.7%	
Chemistry	Harbour College	6	1.17	66.7%	16.7%
Chemistry	Hillcrest College	4	1.25	100.0%	
Chemistry	Lakeside High	6	1.50	100.0%	
Chemistry	Northview College	4	2.00	100.0%	
English	Central Tech	10	1.90	100.0%	20.0%
English	Eastview College	12	1.08	83.3%	
Total		**400**	**1.49**	**80.5%**	**20.8%**